VF Asia Footwear Laboratory Testing Manual

Test Method for Static Coefficient of Friction : James Non-Slip Machine	Test method # VF- B15 Page 1 of 6
Date Issued : June 19, 2010 Revision # :	Created by : Nila Babia Erazo Chris Adam

PURPOSE AND SCOPE: To determine the dry static coefficient of friction of shoe sole or heel materials on controlled walking surfaces and under controlled conditions.

REFERENCES: ASTM F489:96

EQUIPMENT AND MATERIAL:

1. James Non-slip machine meeting the requirements of ASTM F489:96
2. Sample plates : PVC tiles, Mason tiles, Wood Varnish Tiles
3. 400 dry or wet abrasive paper
4. 60 grit abrasive paper
5. Brush
6. Double sided tape or suitable adhesive
7. Distilled water
8. Sample cutter
9. Shoe sole rubber, plastic and EVA.

SAMPLE PREPARATION:

1. Condition the sample for at least 24 hours at the standard laboratory conditions @ 23.0±2.0˚ C Temperature and 55 ±10 % Relative Humidity.
2. Cut specimens with the dimension of 3 “x 3” x 0.25 “from the material to be tested.
3. Remove the finish / outer coat of the specimen using a 60-grit abrasive paper.
4. Then remove the debris of the specimen using the 400-A dry or wet abrasive paper.
5. Brush the sanded specimen to remove the loose particles.
6. Use a dry, clean cloth to remove dust on the sample plate and then put the specimen to the sample plate using the double-faced tape or a suitable adhesive as shown in figures 4 and 5.
7. At least 3 specimens shall be tested.

REPETITIONS: Perform four (4) measurements on all corners per test sample. Eight (8) measurements total per test material for dry and wet testing condition.

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TEST PROCEDURE:

Test should be conducted at standard laboratory conditions @ 23.0±2.0˚C temperature and 55 ±10 % Relative Humidity.

1. Steps to adjust the air pressure (see figure 6):

1.1. Connect the air hose to the air inlet.

1.2. Pull up the air regulator and then turn it clockwise or counterclockwise to obtain a pressure of 5kg/cm^2, which can be read from the pressure meter. After that, push down the air regulator.

2. Sample plate installation:

2.1 Loosen the two setscrews of the test table, then fix the platen to the test table and then lock the setscrews.

2.2 Press the emergency button and the power button and with that press the “Up” button to raise the weight holder to the highest position.

2.3 Press the “Right” button to return the test table automatically.

2.4 Loosen the nut and attach the yoke to the strut. Fix the prepared sample plate to the yoke in such a way that the strut is perpendicular to the locating lever. Then lock the nut (see figures 7 & 8).

3. Testing of Sample:

3.1 Place the weights onto the holder.

3.2 Loosen the screw and move the chart board so that the zero line on the chart is in alignment with the refill of the ball pen. Then lock the screw, see figure 9.

3.3 Press the “Down” button to make the specimen be in firm contact with the test table and in vertical.

3.4 Before the test, pull out the button of the ball pen, otherwise there will not have a curve on the chart, see figure 10.

3.5 Press the “Test” button to start testing.

3.6 In the course of test, the test table moves forward at a rate of 25.4mm/s until the specimen slips and the strut drops.

3.7 Record as the static coefficient of friction the point at which the horizontal curve, made on the chart by the ball pen, changes to a vertical line. If the point is not sharply defined at the top of the

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vertical line, it is necessary to draw a standard curve with the machine. When the specimen stops slipping, the recorded curve at this point is the cure of friction coefficient (see figure 11).

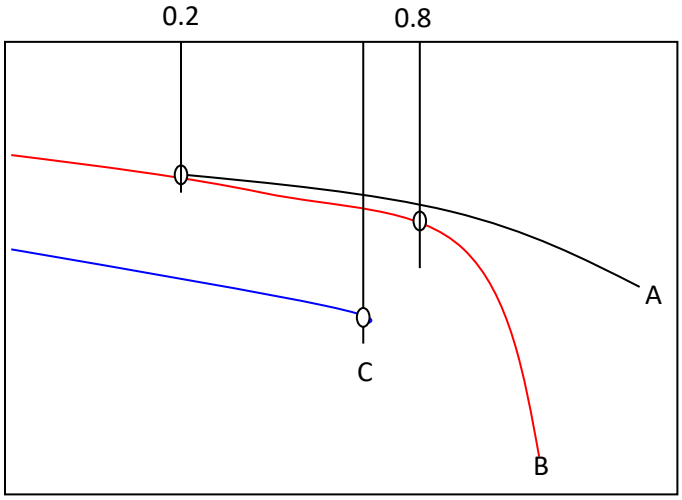


Figure 11

3.8 Chart of coefficient of friction:

3.8.1 At the instant when the specimen slips, immediately step on the foot pedal to allow the refill of the ball pen to withdraw automatically until the strut drops such that a clear curve of friction coefficient appears on the chart, see the curve C in figure 11.

3.9 After the test is complete, the machine stops working. Press the “Up” button to raise the weight holder to the highest position. Press the “Right” button to return the test table automatically.

3.10 Turn off the “Power” and the “Emergency” buttons. Remove the specimen and chart. Read the coefficient of friction from the chart.

REPORT: The average of the 4 coefficient of friction readings to the nearest 0.01kgf/kg.

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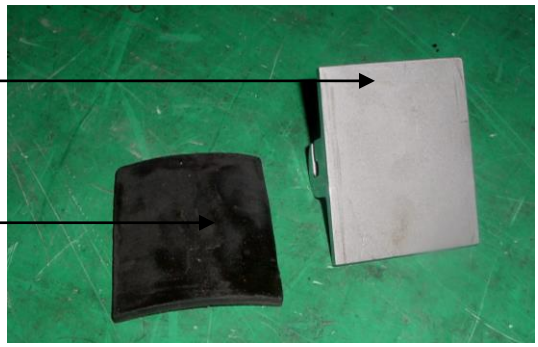


Figure 4

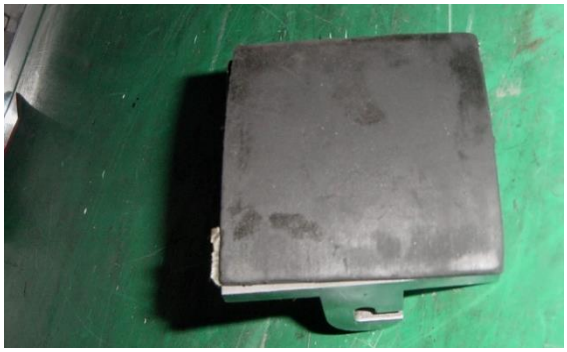


Figure 5

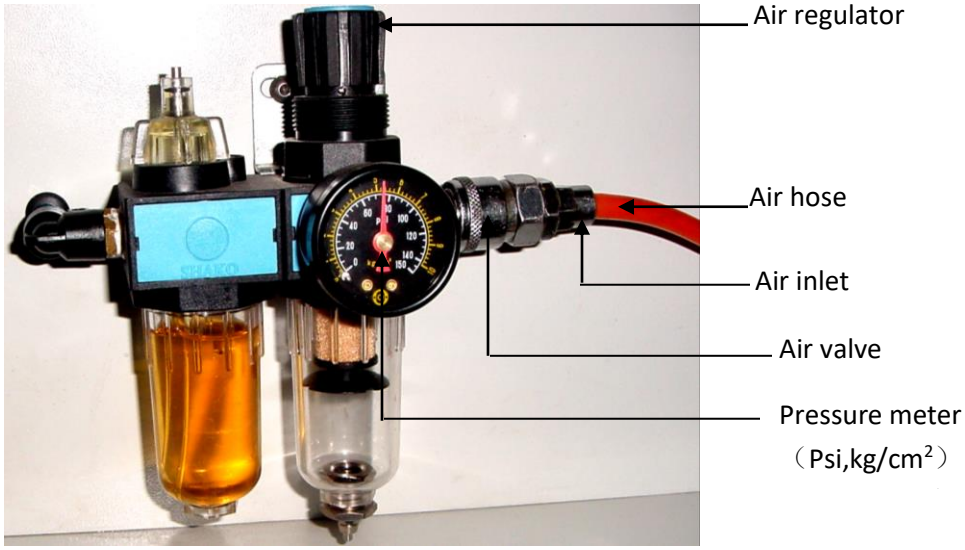


Figure 6

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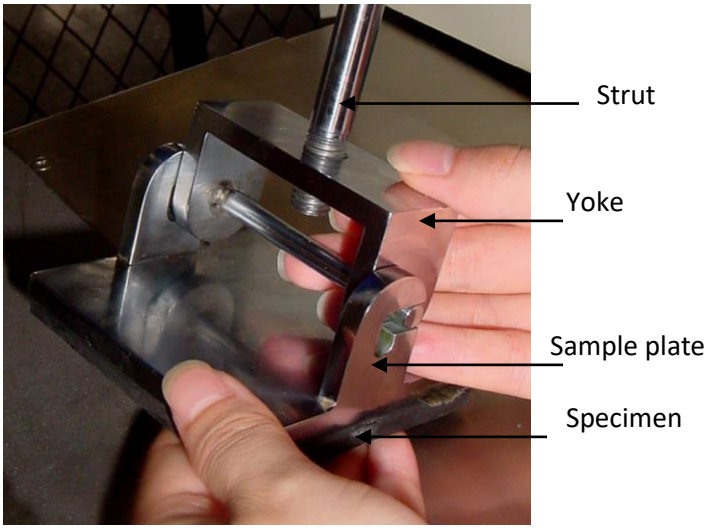


Figure 7

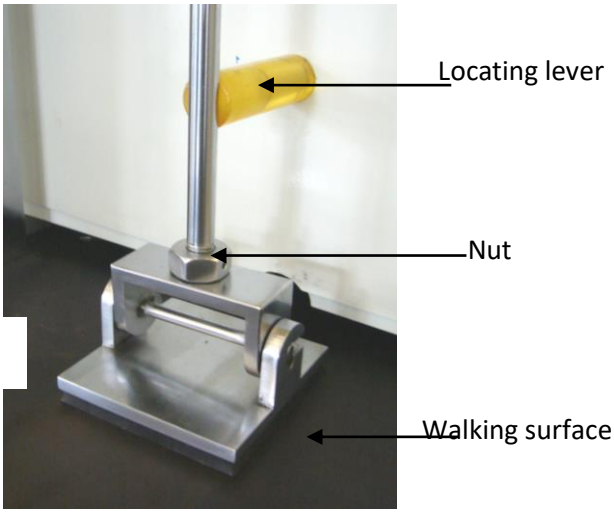


Figure 8

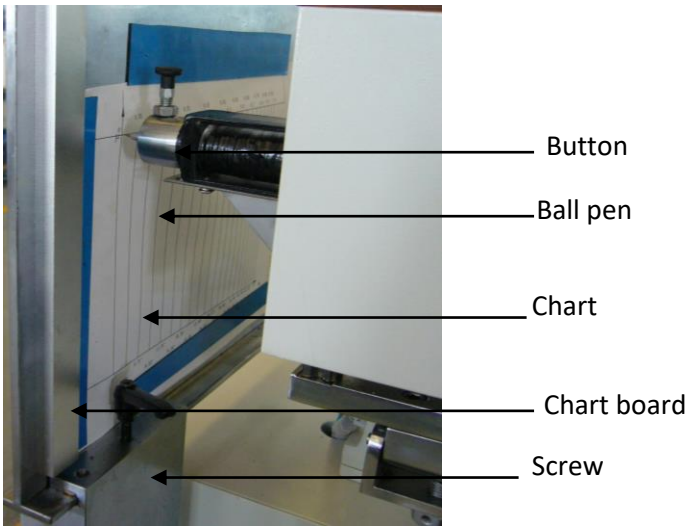


Figure 9



Figure 10

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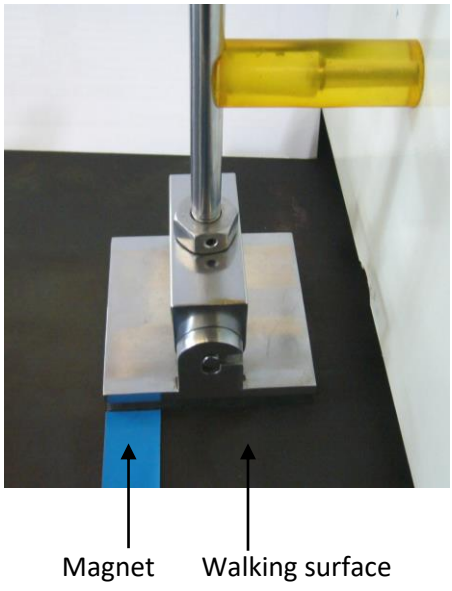


Figure 12

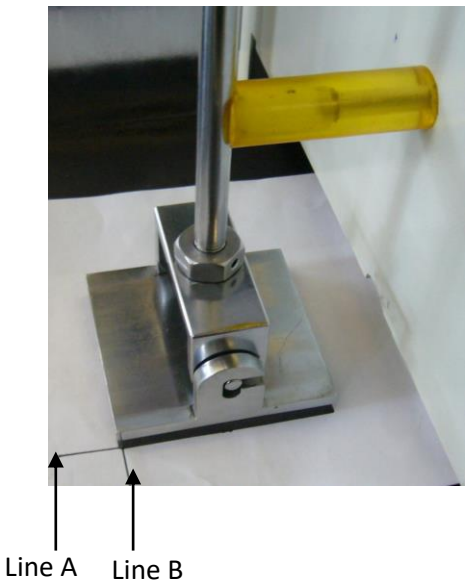


Figure 13